
IC 401P Major Technical Project

Course Code: IC 401P

Course Name: Major Technical Project

L-T-P-C: 0-0-6-4

Prerequisites: Consent of the faculty member

Students intended for: B.Tech

Elective or Core: Core

Approval: 2nd Senate; Credits changed to 0-0-4.5-3 in 6th Senate

Course Contents

In this course, using the information that has been collected and the decisions that have been made about the features, price etc. the teams will either redesign the existing prototype or design a new product based on Market feedback and create the physical product, as well as its packaging. Research at this stage usually involves repeated cycles of product improvement and testing. Product testing includes both physical performance and consumer reactions. The course looks at how a new idea becomes implemented in a system (an organization or society) and the factors that influence the adoption of a new idea. This course also looks at the influence of individuals and groups within the change process and how they affect the acceptance of new ideas. Finally, the course explores the prediction and consequences of new technologies.

The end-result of these efforts is a product that meaningfully adds value to the students, faculty and society.

There may be few teams working on specific components or sub systems or technology.

Key Learning Topics

Engineering drawing (CAD) of modified product, Detailed design, Design of experiments, Multiobjective design optimization. CAE Analysis, Software development, Design for assembly, Design for manufacturing, Product manufacturing.: CAM programming, operation of CNC machining equipment and rapid prototyping. Open House

References:

1. Geoffrey Boothroyd , Peter Dewhurst , Winston A. Knight, **Product Design for Manufacture & Assembly**, 2nd Edition.
2. Kathleen Fasanella, **The Entrepreneur's Guide to Sewn Product Manufacturing**.